Step 2: Organize and Describe the Data

The second step of the project aims at defining and structuring all essential data necessary for the automated pet feeding system. It lists all required inputs, including components such as the real-time clock, food level sensor, and weight sensor placed beneath the bowl. The expected results of the system are also taken into account. To ensure the system operates smoothly and effectively, sample values and operational limitations are duly anticipated. In short, this step outlines inputs and outputs with their purpose, format, and operational specifications in the comprehensive data table below.

Component	Type	Purpose/ description	Data Format/ signal	Operational Parameters
Feeding Schedule	Input	Preset the time when the feeder should dispense food	Time value (HH:MM)	1-3 feeding times per day (specifically will be based on the shelter or user needs)
The real time clock	Input	Track the current time to match with the schedule	Digital timer	The data need to be accurate second by second
Food sensor	Input	Users use it to detect food level or if there is any food currently	Digital or Analog (in gram)	Threshold <= 20g = “not eaten” → which can measure the amount of food that has been consumed
Servo Motor	Output	Dispense food via a mechanical rotation	PWM signal (a tool for controlling electronic devices)	Be able to rotate to release the portion of food out at scheduled time
Alert system	Output	Notify users of the issue (via sound, light, messages,...)	LED/ Buzzer/ SMS	Activate after 10-15 minutes when the issue still persists

Status display	Output	Show real-time status, errors and the next feeding time	Alphanumeric (ASCII characters) For example: “`Next Feed: 18:00"`	Update every second, display feeding history for the last 24 hours
Manual button	Input	Allow users to manually trigger the food dispensing or reset the feeder in case any issue occurs	Digital input signal (ON/OFF)	Short press: dispensing Long press: system reset